

INGREDIENT SCIENCE

DIY FORMULAS

INTRA-WORKOUT

RECOVERY STACK

PRE · INTRA · POST WORKOUT OPTIMIZATION COMPLETE SUPPLEMENT GUIDE

Every Ingredient Analyzed · DIY Formulas · Intra-Workout Fuel · Recovery Stack

Stop wasting money on proprietary blends — understand what actually works

ELITE FITNESS GUIDE SERIES

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■ 1. PRE-WORKOUT SCIENCE

The pre-workout supplement market is a ■5,000 crore industry globally, yet most products contain proprietary blends with clinically ineffective doses behind exotic names. This guide cuts through the marketing to show you **exactly which ingredients work, at what doses, and why.**

■ A 2021 systematic review (Martinez et al., *Journal of the International Society of Sports Nutrition*) analyzed 44 commercially available pre-workout products. Finding: 75% were under-dosed for key active ingredients relative to clinically validated doses. The average product contained ingredients at 40% of the evidence-based dose.

Evidence Level	Ingredients	Benefit
Grade A (Very Strong)	Caffeine, Creatine, Citrulline, Beta-Alanine	Primary drivers of pre-workout performance
Grade B (Good)	Betaine, Alpha-GPC, Tyrosine, Theanine	Meaningful secondary contributors
Grade C (Moderate)	Beet Root Extract, Norvaline, Taurine	Modest benefits; worth including at correct dose
Grade D (Weak/Inconclusive)	BCAAs pre-workout, Glutamine, Vitamin B12	Limited evidence for acute pre-workout benefit
Grade F (No Evidence)	'Proprietary blends', fairy-dusted exotics, most herbals at under-studied doses	Marketing only

■ 2. THE STIMULANT LAYER

Caffeine Anhydrous — The Gold Standard Stimulant:

Caffeine is the most researched ergogenic aid in sports nutrition history. It works via adenosine receptor antagonism — blocking the sleep-inducing signal in your brain. This increases adrenaline, dopamine, and acetylcholine simultaneously.

- **Optimal dose:** 3–6 mg/kg body weight. 70kg athlete: 210–420 mg
- **Timing:** 30–45 min before training (peak plasma: 45–60 min)
- **Tolerance management:** Cycle caffeine — take 2 days off per week to prevent receptor downregulation
- **Benefits:** 10–15% increase in strength output; significant endurance improvement; pain perception reduction; improved focus and motivation
- **Caution:** Half-life 5–6 hours; avoid after 2 PM if bedtime is 10 PM

Dynamine (Methylliberine) + TeaCrine (Theacrine):

- Dynamine: Faster onset than caffeine (15 min); dopaminergic; pairs well with caffeine
- TeaCrine: Same mechanism as caffeine but zero tolerance development over time
- Effective stack: Caffeine 200mg + Dynamine 100mg + TeaCrine 50mg — fast onset + sustained effect

■ 3–4. THE PUMP & PERFORMANCE LAYERS

Ingredient	Clinical Dose	Mechanism	Effect
L-Citrulline	6–10 g (not citrulline malate)	Converts to arginine in kidneys → nitric oxide (NO) production → vasodilation	Best pump ingredient; superior to arginine due to better oral bioavailability
Agmatine Sulfate	500 mg–1 g	NOS inhibitor → amplifies NO signaling; also reduces pain perception	Synergistic with citrulline; unique pain-reducing benefit
Norvaline	100–200 mg	Arginase enzyme inhibitor — prevents breakdown of arginine into urea; prolongs NO effect	Modest addition; dose-dependent benefit
HydroMax Glycerol	2–3 g	Hyperhydrates muscle cells; draws water into muscle → skin-splitting pump; anti-dehydration	Excellent for appearance and performance in hot conditions

Ingredient	Clinical Dose	Mechanism	Expected Benefit
Beta-Alanine	3.2–6.4 g	Combines with histidine → carnosine in muscle → buffers hydrogen ions (acid) → delays fatigue	~3% endurance increase; more reps in 8–20 rep range. Causes tingles (paraesthesia) — normal, harmless
Creatine Monohydrate	3–5 g (maintenance) or 20g/day × 5 days (loading)	Increases phosphocreatine stores → faster ATP regeneration → more power and strength	3–8% strength increase; 1–3 kg lean mass over 4 weeks; one of the most proven supplements ever
Betaine Anhydrous	2.5 g	Osmolyte — improves cellular hydration similar to creatine; may increase GH and IGF-1	~4% strength; 2% power output increase; 1.3 kg lean mass in some trials

■ 5–6. FOCUS LAYER & DIY FORMULA

Ingredient	Dose	Mechanism	Best For
Alpha-GPC	300–600 mg	Acetylcholine precursor — enhances mind-muscle connection; increases GH output acutely	Mind-muscle connection; neurological focus
L-Tyrosine	500–2000 mg	Dopamine and noradrenaline precursor; maintains cognitive performance under stress/fatigue	Focus under high-stress training; stimulant augmentation
L-Theanine	100–200 mg (with caffeine)	GABAergic modulation; reduces caffeine jitteriness; improves 'calm focus'	Pairs with caffeine 2:1 ratio; reduces anxiety without reducing stimulation
Huperzine A	50–200 mcg	Acetylcholinesterase inhibitor — prevents breakdown of acetylcholine; long half-life	Occasional use only (every 2–3 days) — accumulates; avoid daily

DIY Pre-Workout Formula (per serving):

Ingredient	Amount	Cost/Serving
Caffeine Anhydrous	200 mg	■0.50
L-Citrulline	6000 mg	■8–12
Beta-Alanine	3200 mg	■4–6
Creatine Monohydrate	5000 mg	■3–5
Alpha-GPC 50%	300 mg	■6–10
L-Theanine	200 mg	■2–4
Betaine Anhydrous	2500 mg	■5–8
Agmatine Sulfate	500 mg	■3–5
TOTAL	~17.7g powder	■30–50 (vs ■200–500 for commercial)

■ 7–8. INTRA-WORKOUT & POST-WORKOUT

Intra-Workout Nutrition (for sessions >60 minutes):

Component	Amount	Source	Purpose
Fast Carbohydrates	20–40 g per hour	Dextrose, Cyclic Dextrin, banana, dates	Maintains muscle glycogen; prevents cortisol spike; sustains performance in 2nd hour
Essential Amino Acids (EAAs)	6–10 g	EAA supplement or 20g whey (smaller sip)	Prevents muscle protein breakdown; stimulates synthesis during training
Electrolytes	Full spectrum	Sports drink / electrolyte powder / salt + potassium	Prevents hyponatremia; maintains muscle contraction; crucial in Indian heat
Water	500–800 mL per hour	Plain water	Prevents dehydration-induced strength loss (2% dehydration = 10% strength loss)

The Anabolic Window — Myth vs Reality:

■ Aragon & Schoenfeld (2013, JISSN) meta-analysis: the 'anabolic window' of 30–45 minutes post-workout is largely a myth for athletes who consumed a pre-workout meal. Muscle protein synthesis is elevated for 24–48 hours post-training — total daily protein intake matters far more than timing.

- **If fasted before training:** Post-workout meal is critical — consume protein + carbs within 30 min
- **If fed before training:** Post-workout meal within 2–3 hours is sufficient
- **Always prioritize:** Total daily protein (1.5–2g/lb) over meal timing

Post-Workout Supplement Stack:

Supplement	Dose	Timing	Evidence
Whey Protein Isolate	25–40 g	Within 1 hr post-workout	Strong — fastest digesting complete protein; leucine content drives MPS
Creatine Monohydrate	5 g	Post-workout (slightly better than pre)	Strong — marginally superior post-workout absorption in one meta-analysis
Fast Carbohydrates	40–80 g	Post-workout with protein	Strong — restores glycogen; insulin spike aids creatine uptake
Tart Cherry Extract	480 mg	Post-workout	Moderate — reduces muscle soreness and inflammation significantly

Knowledge of ingredient mechanisms eliminates expensive marketing decisions. Build your stack on science, not branding.